JC3510: Intelligent Software Implementation (2025-2026)

Assessment: Group Project

(This assessment accounts for 50% the total mark of the course)

Course Code	JC3510
Academic Year	2025-26
Session	Second Half Session
Study Programme	Undergraduate
Level	3
Credit Points	30 Credits (15 ECTS)

This assessment is the second of two coursework assignments and involves collaborative, team-based work, supervised by an assigned academic.

Learning Outcomes

On successful completion of this assignment, the student will demonstrate the ability to:

- Apply effective written and verbal communication skills.
- Analyse and apply current software design trends and practices.
- Create a software-centric system.
- Apply project management and teamwork skills.
- Understand professional practices in software development.

Overview

The team project, a cornerstone of this course, is structured around several key components designed to mirror the stages of real-world software development. These components are:

- Project Proposal (PASS/FAIL): A comprehensive plan outlining the project idea, target users, scope, technical specifications, development methodology, timeline, and required resources.
- Final Report (60%): A detailed technical document describing the system design, development process, implementation decisions, and evaluation of outcomes.
- Proof of Concept (PoC) Software (40%): A functional software artifact demonstrating the practical application of software design principles and development techniques learned in the course.
- Presentation (PASS/FAIL): A formal presentation in which teams showcase their project, explain design decisions, and discuss achieved results.

Detailed instructions for each component are provided in subsequent sections, guiding you through the expectations for each stage of the project.

❖ Project Proposal Instructions (PASS/FAIL)

The Project Proposal is a critical component of your team project. It defines the problem space and planned solution. It should clearly articulate what the team intends to build and how the work will be approached.

The proposal must include:

- A clear and well-motivated project idea, including the problem being addressed and its relevance.
- Identification of target users and their primary needs.
- A defined project scope, including core features and explicit exclusions.
- High-level technical specifications, such as target platform, core technologies, and high-level architectural approach.
- The chosen development methodology and justification for its suitability.
- A realistic timeline with key milestones.
- An overview of required resources, including development tools, external libraries, services, and team roles.

The proposal should demonstrate feasibility, coherence, and alignment with the course objectives.

<u>Pass/Fail Outcome and Penalty Policy</u>

- The Project Proposal is assessed on a **Pass/Fail** basis and is a **mandatory component** of the team project.
- A **Pass** indicates that the proposed project is feasible, coherent, and appropriately scoped, and that it aligns with the course objectives.
- A **Fail** indicates that the proposal is incomplete, infeasible, unclear, or significantly misaligned with the course objectives.
- Teams that receive a Fail will be given one opportunity to revise and resubmit the proposal.
- If the proposal is not passed after resubmission, **a 5% deduction** will be applied to the total project score (calculated from the Final Report and Proof of Concept Software).

<u>Format and Submission</u>

- Length: 5 to 7 pages.
- Format: No specific template; use a legible font and standard margins.
- Submission: The proposal should be submitted in PDF format using the appropriate submission point on Blackboard.

❖ Final Report Instructions (60%)

The final report is a comprehensive document that encapsulates the entirety of your team project, from conception to completion. It provides an opportunity to reflect on the project process, document the work performed, and analyse the outcomes.

The recommended structure of the final report:

- Introduction and System Overview: Introduces the problem being addressed and presents a high-level overview of the resulting system, including its purpose, main capabilities, and operating context.
- System Design: Describes the system architecture, major components, their responsibilities, interactions, and the rationale for key system-level design decisions.
- Implementation: Describes how the system design was realized in code, including the technology stack, implementation-specific decisions, application of design principles and patterns, and significant technical challenges.
- Development Process: Explains the development approach, iteration strategy, milestone progression, and adaptations made during the project lifecycle.
- Evaluation of Outcomes: Assesses how well the implemented system meets its stated objectives, highlighting strengths, limitations, and supporting evidence.
- Conclusion and Future Work: Summarizes key design insights and identifies realistic improvements or extensions based on project outcomes.

If required, a list of references (maximum of 20) cited in the report may be included. The source code must not be included in the report.

Evaluation Criteria

- Completeness and clarity of the information provided
- Depth of analysis and quality of reflection on the project process and outcomes
- Quality of technical design, implementation, and evaluation
- Adherence to the specified formatting guidelines
- Overall coherence and professionalism of the document

Format and Submission

- Length: 20 to 30 pages
- Format: Readable font (e.g., Times New Roman, 12-point), single-spaced, single column, with numbered pages and section headings
- Submission: PDF format via the appropriate submission point on Blackboard

❖ Proof-of-Concept (PoC) Software Instructions (40%)

The PoC demonstrates the practical realization of the proposed design. It does not need to be a complete product but must be technically credible. Alongside, include a detailed user manual for guidance on installation, operation, and utilization of your program.

The PoC must:

- Implement the core functionality defined in the approved proposal
- Reflect the stated architecture and design principles
- Be stable enough to demonstrate intended behaviour

The PoC should convincingly show that the proposed design can be implemented and extended.

<u>User Manual Guidelines</u>

- Content: Provide comprehensive instructions for installing, running, and using your program. Highlight any prerequisites or configurations needed. Include GitHub Repo link if available.
- Length: 10-15 pages.
- Format: PDF file.

<u>Evaluation Criteria</u>

- Demonstration of the core functionality and viability of the project idea
- Technical quality of the PoC, including architectural alignment, stability, and extensibility
- Clarity, completeness, and professionalism of the user manual
- Adherence to the submission guidelines and instructions

<u>Submission</u>

- Include your sources code, other necessary files, and the PDF user manual into a single zip file.
- Name the zip file clearly with your group number for easy identification.
- Submit the zip file using the appropriate submission point on Blackboard.

❖ Presentation Instructions (PASS/FAIL)

The presentation evaluates the team's ability to communicate technical ideas effectively and defend system design, implementation, and development choices.

The presentation should:

- Clearly introduce the problem, objectives, and target users
- Summarize the system design and key architectural decisions
- Demonstrate the PoC and its main features
- Highlight significant challenges, decisions, and lessons learned
- Conclude with an assessment of results and potential future improvements

All team members are expected to participate meaningfully.

Format and Submission

- Slides: Each team must prepare a set of presentation slides.
- Recorded Presentation: Each team must submit a recorded video of the presentation, in which team members present the slides and explain the project.
- Recorded Video Duration: 10-15 minutes.
- Submission: Both the slide deck and the recorded video must be submitted via the appropriate submission point on Blackboard.

Pass/Fail Outcome and Penalty Policy

- The Presentation is assessed on a **Pass/Fail** basis and is a **mandatory component** of the team project.
- A **Pass** requires clear communication of project objectives, system design and implementation, successful demonstration of the Proof of Concept, and meaningful participation by all team members.
- A **Fail** is assigned if these requirements are not met.
- If the Presentation is not passed, a 5% deduction will be applied to the total project score (calculated from the Final Report and Proof of Concept Software).

Group Workload Profile Instructions

The Group Workload Profile serves as a transparent record of each member's contribution, both quantitatively and qualitatively, throughout the project duration.

Process

- Recording Effort: Each group member must document their tasks, responsibilities, and contributions to the project. This includes both the amount of work done (quantitative) and the quality or impact of that work (qualitative).
- Open Agreement: Team members must collaboratively discuss and agree on the contribution level of each member. This discussion should be open, fair, and based on documented efforts to ensure an accurate assessment of everyone's input.
- Contribution Assessment: Contributions can be categorized and assessed in various aspects, such as project planning, execution, problem-solving, creativity, and leadership roles, among others.

Completion and Submission

- A designated form or template will be made available to all groups for submitting their workload profiles.
- The specific details on how to complete the workload profile, including the format and criteria for assessment, will be provided.
- Submit the completed form using the appropriate submission point on Blackboard.

Special Note: While the group workload profile is an important consideration, it may not be applied if it leads to an unfair distribution of scores across groups. In such cases, the group workload profile will not be factored into the final grade for any group.

❖ Other Guidelines and Information

Group Formation and Supervision

At the start of the course, groups will be formed, and each will be assigned a supervisor. The group leader is responsible for reaching out to their assigned supervisor to introduce the team. Teams are encouraged to schedule meetings with their supervisor and receive guidance throughout the project. Supervisor allocation will be communicated through Blackboard notices.

Important Dates

Project Proposal: 30 March 2026 Monday (23:59 China Standard Time).

Final Report: 01 June 2026 Monday (23:59 China Standard Time).

PoC Software: 08 June 2026 Monday (23:59 China Standard Time).

Presentation: 15 June 2026 Monday (23:59 China Standard Time).

Group Workload Profile: 16 June 2026 Tuesday (23:59 China Standard Time).

All dates provided are provisional and may be adjusted based on circumstances. Any changes will be communicated through the Blackboard noticeboard.

Information for Plagiarism and Conduct

Your submitted report may be submitted for plagiarism check (e.g., Turnitin). Note that GenAI tools such as ChatGPT to generate material for work submitted for assessment, without any form of editing or acknowledgement, will be investigated as a plagiarism offence. Please also read the following guide developed by the university to understand what academic misconduct means, how you can avoid it, and what the penalties are should you be found to be responsible for committing academic misconduct.

https://www.abdn.ac.uk/students/academic-life/academic-integrity.php

Use of AI-Assisted Tools

The use of AI-assisted tools is permitted **only as a supplementary aid** and must not replace a team's own analysis, design work, or implementation effort. Any use of such tools **must be explicitly acknowledged** in the relevant project deliverables.

AI tools should be used responsibly and only where they clearly add value. The use of AI must not undermine individual or team understanding, learning outcomes, or academic integrity. Teams remain fully responsible for all submitted work.

Late Submission

Unauthorized late submissions may incur penalties, which will substantially reduce your overall CGS points, according to the university policy. You are advised to check the latest policy in this regard, available in the Course Information folder on MyAberdeen (Blackboard).

Questions

Any questions pertaining to any aspects of this assessment, please address them to the course coordinator Dr. Tryphon Lambrou via Blackboard's Message function available on MyAberdeen's course site or via email tryphon.lambrou@abdn.ac.uk